Type-k Neutrosophic Sets: Recursive Triadic Structures and Their Empirical Realization in Large Language Models

Florentin Smarandache[1]
Maikel Leyva-Vázquez[2,3,4]

[1]Department of Mathematics, University of New Mexico, Gallup, NM 87301, USA. smarand@unm.edu
[2]Facultad de Ciencias Matemáticas y Físicas, Universidad de Guayaquil, Guayaquil, Ecuador.
[3]Universidad Bolivariana del Ecuador, Duran, Ecuador.
[4]Universidad Bernardo O'Higgins, Santiago, Chile.
maikel.leyvavazquez@ug.edu.ec

May 2026

Abstract

We introduce *Type-k Neutrosophic Sets*, a recursive extension of neutrosophic logic in which each epistemic component (T, I, F) is itself characterised by a full neutrosophic triplet, yielding 3^k scalars per element at recursion depth k. We prove a strict expressive hierarchy Type-1 $\subsetneq$ Type-2 $\subsetneq \cdots \subsetneq$ Type-k and identify a canonical embedding φ that maps any Type-$(k-1)$ element into its Type-k representation. We propose a recursive aggregation operator that reduces to the standard Single-Valued Neutrosophic Weighted Average (SVNWA) at $k = 1$. Empirically, we demonstrate that large language models (LLMs) implicitly compute Type-2 values when evaluation protocols permit components outside $[0, 1]$: across 2,419 evaluations of six foundation models under five epistemic protocols, 24.7% of all responses require Type-2 representation. Under the overset protocol S4-O.A, this fraction rises to 97.3%, and is consistent across all six vendors (19–30%). The restriction to Type-1 is a *protocol artefact*, not an epistemic limit. Type-k Neutrosophic Sets provide the first formal framework that makes these LLM behaviours mathematically legitimate.

Keywords: neutrosophic sets; type-2 neutrosophic; recursive neutrosophic; large language models; epistemic uncertainty; epistemic indeterminacy; foundation models.

1 Introduction

Smarandache's neutrosophic logic [1] represents propositions as triplets (T, I, F) where T is the degree of truth, I the degree of indeterminacy, and F the degree of falsity, with $T, I, F \in [0, 1]$ independently (no sum constraint). This *Type-1* representation has found application in multi-criteria decision making [5], image processing, and, more recently, in the epistemic evaluation of large language models (LLMs) [8, 9].

A fundamental question arises naturally: if T is the degree to which a proposition is true, to what degree is T *itself* true, indeterminate, or false? In type-2 fuzzy logic [7], membership degrees are themselves fuzzy sets, enabling a second layer of uncertainty. The neutrosophic analogue has not been formally defined.

In this paper we fill that gap. We introduce *Type-k Neutrosophic Sets*, where each component of a Type-$(k-1)$ representation is replaced by a full neutrosophic triplet. The resulting structure has 3^k scalars per element at depth k and subsumes Type-1 as the base case.

Our motivation is both theoretical and empirical. On the theoretical side, the construction is the most natural extension of neutrosophic logic to nested uncertainty. On the empirical side, we show that state-of-the-art LLMs *already compute Type-2 values* when the evaluation protocol opens the range beyond $[0, 1]$: negative T values ("anti-truth"), $I > 1$ ("hyper-indeterminacy"), and $F > 1$ ("hyper-falsity") appear in nearly all vendors under the appropriate protocol. These values are mathematically inconsistent in Type-1 but are precisely representable in Type-2.

The rest of the paper is organised as follows. Section 2 reviews Type-1 neutrosophic sets and existing extensions. Section 3 introduces Type-k Neutrosophic Sets (Definitions 1–3) and states the strict hierarchy theorem. Section 4 defines the recursive aggregation operator. Section 5 presents the empirical evidence from LLM evaluations. Section 6 connects the construction to existing neutrosophic and plithogenic frameworks. Section 7 concludes.

2 Background

2.1 Type-1 Neutrosophic Sets

Definition 1 (Type-1 Neutrosophic Set, [1]). *Let U be a universe of discourse. A* neutrosophic set $A^{(1)}$ *on U assigns to each element $x \in U$ a triplet*

$$\mu^{(1)}(x) = (T, I, F), \quad T, I, F \in [0, 1],$$

where T, I, F are independent: there is no constraint $T + I + F = 1$. The special case $T + I + F \leq 1$ recovers the intuitionistic fuzzy set of Atanassov [6].

2.2 Existing extensions

Several extensions of Type-1 have been proposed. *Interval-valued neutrosophic sets* [5] allow T, I, F to be closed intervals in $[0, 1]$. *Refined neutrosophic logic* [2] splits each component into sub-components: $T \to (T_1, \ldots, T_p)$, $I \to (I_1, \ldots, I_r)$, $F \to (F_1, \ldots, F_s)$. These sub-components remain scalars. *Neutrosophic overset / underset / offset* [3] allows components outside $[0, 1]$: $T, I, F \in [-\Omega, \Omega^+]$ for bounds $\Omega, \Omega^+ > 0$. *Plithogenic sets* [4] assign a contradiction degree between attribute values and extend the aggregation accordingly.

None of these extensions assign a *full neutrosophic triplet* to each component, which is the defining feature of Type-k Neutrosophic Sets.

3 Type-k Neutrosophic Sets

3.1 Definitions

Definition 2 (Type-2 Neutrosophic Set). *Let U be a universe of discourse. A* Type-2 Neutrosophic Set $A^{(2)}$ *on U assigns to each $x \in U$ nine scalars organised as three nested triplets:*

$$\mu^{(2)}(x) = \Big(\underbrace{(T_T,\, I_T,\, F_T)}_{\textit{triplet of } T},\ \underbrace{(T_I,\, I_I,\, F_I)}_{\textit{triplet of } I},\ \underbrace{(T_F,\, I_F,\, F_F)}_{\textit{triplet of } F}\Big), \tag{1}$$

where all nine scalars are in $[0, 1]$ independently. The semantics *of each sub-triplet is:*

- (T_T, I_T, F_T)*: to what degree is T itself true / indeterminate / false.*
- (T_I, I_I, F_I)*: to what degree is I itself true / indeterminate / false.*
- (T_F, I_F, F_F)*: to what degree is F itself true / indeterminate / false.*

Definition 3 (Type-3 Neutrosophic Set). *A* Type-3 Neutrosophic Set $A^{(3)}$ *on U assigns to each $x \in U$ twenty-seven scalars. Each of the nine scalars of a Type-2 representation is itself replaced by a neutrosophic triplet. For the T component:*

$$T \to \Big(T_T(T_{TT}, I_{TT}, F_{TT}),\ I_T(T_{IT}, I_{IT}, F_{IT}),\ F_T(T_{FT}, I_{FT}, F_{FT})\Big), \tag{2}$$

and analogously for I and F.

Definition 4 (Type-k Neutrosophic Set). *Let $k \geq 1$ be an integer. A* Type-k Neutrosophic Set $A^{(k)}$ *on U is defined recursively:*

- Base case *($k = 1$): $A^{(1)}$ is a Type-1 neutrosophic set (Definition 1).*
- Recursive case *($k \geq 2$): $A^{(k)}$ is obtained from $A^{(k-1)}$ by replacing each scalar component with a full neutrosophic triplet of scalars in* $[0, 1]$.

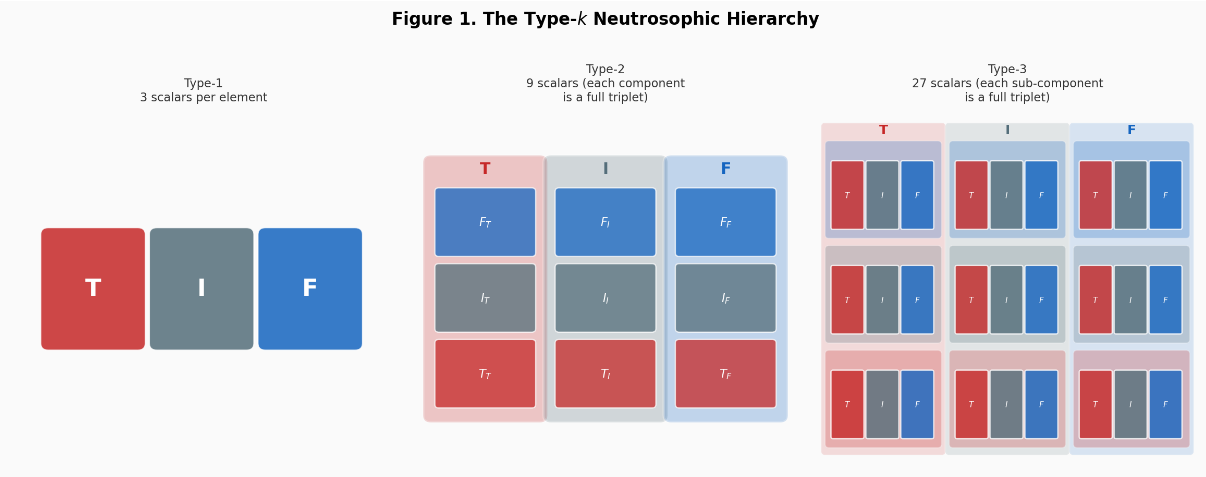

Figure 1: **Figure 1.** The Type-k neutrosophic hierarchy. **Left (Type-1):** three independent scalars (T, I, F). **Centre (Type-2):** each component is itself a neutrosophic triplet, yielding 9 scalars. **Right (Type-3):** each sub-component is recursively a triplet, yielding 27 scalars. At depth k, an element requires 3^k scalars.

Each element $x \in U$ is characterised by 3^k scalars. The set of Type-k neutrosophic sets over U is denoted $\mathcal{NS}^{(k)}(U)$.

Definition 5 (Subset-valued Neutrosophic Set)**.** *In the most general construction, proposed by Smarandache (personal communication, 2026-05-04), the components T, I, F of a neutrosophic element are not scalars or finite recursive triplets but* neutrosophic sets *over a sub-universe U':*

$$T,\ I,\ F\ \in\ \mathcal{NS}^{(1)}(U').$$

This admits variations in what each epistemic category means*: different sub-elements of U' can represent different modes of truth, indeterminacy, or falsity, with their own internal neutrosophic structure. Every finite Type-k set is a special case obtained by taking U' to be a finite set of cardinality 3^{k-1} and fixing the neutrosophic memberships of its elements.*

Corollary 1 (Ultimate hierarchy)**.** *The chain $\mathcal{NS}^{(1)} \subsetneq \mathcal{NS}^{(2)} \subsetneq \cdots \subsetneq \mathcal{NS}^{(k)} \subsetneq \cdots \subsetneq \mathcal{NS}^{(\text{subset})}$ is strict at every level, where $\mathcal{NS}^{(\text{subset})}$ denotes the class of subset-valued neutrosophic sets.*

Remark 1. *The subset-valued construction (Definition 5) is the most expressive framework: it allows not only* how much *truth is present but also* what kind *of truth, leaving room for interpretive variation across contexts, cultures, or epistemic communities. The formal development of aggregation, distance, and decision operators under Definition 5 is left to future work.*

3.2 Canonical embedding

Definition 6 (Embedding φ)**.** *The canonical embedding $\varphi : \mathcal{NS}^{(k-1)}(U) \to \mathcal{NS}^{(k)}(U)$ maps each scalar s in a Type-$(k-1)$ element to the Type-k triplet $(s, 0, 1-s)$, i.e., a certain scalar is embedded as a triplet with zero indeterminacy and complementary falsity.*

3.3 Expressive hierarchy

Theorem 1 (Strict hierarchy)**.** *$\mathcal{NS}^{(1)}(U) \subsetneq \mathcal{NS}^{(2)}(U) \subsetneq \cdots \subsetneq \mathcal{NS}^{(k)}(U)$ in expressive power.*

Proof. Inclusion $\subseteq$: every Type-$(k-1)$ element x embeds into Type-k via φ (Definition 4).

Strictness $\subsetneq$ (witness for $k = 2$): Consider the proposition "This statement is both true and false" evaluated by a language model under an epistemic protocol that allows values in $[-1, 2]$. Suppose the model returns $\mu_T^* = -0.20$, meaning "the truth of this claim is more false than true" (anti-truth regime). In Type-1, we require $T \in [0, 1]$, so $\mu_T^* = -0.20$ is *inadmissible*. In Type-2, the same state is admissible as:

$$(T_T, I_T, F_T) = (0.00,\ 0.20,\ 0.80),$$

meaning T is 0% true, 20% indeterminate, and 80% false — a well-formed Type-2 triplet with all components in $[0, 1]$. Empirically, we observe 65 such anti-truth responses (2.7% of evaluations; see Section 5). The same argument generalises to the witness $(T = 1.7, I = 1.8, F = 1.6)$ (Table 4), which is inadmissible in any Type-1 extension with $T, I, F \in [0, 1]$ but fully representable in Type-2. $\square$

4 Recursive Aggregation Operator

The standard SVNWA operator [5] at Type-1:

$$\mathrm{SVNWA}(\mu_1, \ldots, \mu_n; \mathbf{w}) = \Big(1 - \prod_{i=1}^{n}(1 - T_i)^{w_i},\ \prod_{i=1}^{n} I_i^{w_i},\ \prod_{i=1}^{n} F_i^{w_i}\Big).$$

Definition 7 (Type-k SVNWA)**.** *The* Type-k SVNWA *applies the SVNWA operator recursively to each sub-triplet independently at every level. For Type-2 with weight vector* $\mathbf{w}$*:*

$$SVNWA^{(2)}(\mu_1^{(2)}, \ldots, \mu_n^{(2)}; \mathbf{w}) = \Big(SVNWA(T_1, \ldots, T_n; \mathbf{w}),\ SVNWA(I_1, \ldots, I_n; \mathbf{w}),\ SVNWA(F_1, \ldots, F_n; \mathbf{w})\Big), \quad (3)$$

where each argument of the outer tuple is itself the SVNWA of the corresponding sub-triplets across elements $1, \ldots, n$*.*

Proposition 1 (Reduction)**.** *$SVNWA^{(k)}$ reduces to the standard SVNWA when all Type-k elements are images of Type-1 elements under φ (i.e., when all sub-triplets have $I = 0$ and $F = 1 - T$).*

Proof. Under the embedding φ, each sub-triplet $(s, 0, 1 - s)$ has $I = 0$, so $\prod I_i^{w_i} = 0$, and $F = 1 - T$ recovers the standard complement. The SVNWA of the T-sub-triplets then equals the standard SVNWA of the original T values. $\square$

5 Empirical Evidence: LLMs as Implicit Type-2 Computers

5.1 Experimental design

We reanalysed 2,419 evaluations from a multi-vendor empirical study [9]. Six foundation models were evaluated on five epistemic phenomena (Paradox, Ignorance, Vagueness, Contradiction, Contingency) under five protocols:

- S1: classical single scalar $T \in [0, 1]$.
- S4: Mason's [8] three-component declared-loss frame $(what, why, severity)$.
- S4-N: per-attribute neutrosophic (T, I, F) triplet.
- S4-O.A: overset protocol with $T, I, F \in [0, 2]$.
- S4-O.C: offset protocol with $T, I, F \in [-1, 2]$.

The six vendors were: Alibaba (Qwen-3-235B), Anthropic (Claude Sonnet 4), DeepSeek Chat, Meta (Llama-4-Maverick), Mistral Medium-3.1, and OpenAI (GPT-4o).

5.2 Type-2 identification criterion

A response $\mu = (T, I, F)$ *requires Type-2 representation* if and only if at least one component falls outside $[0, 1]$: $T < 0$, $T > 1$, $I > 1$, or $F > 1$. Such responses are inadmissible in any Type-1 neutrosophic set (including the overset/offset extensions of [3] when the extended range is *not* assumed) but are representable in Type-2 via the explicit mapping defined in the proof of Theorem 1.

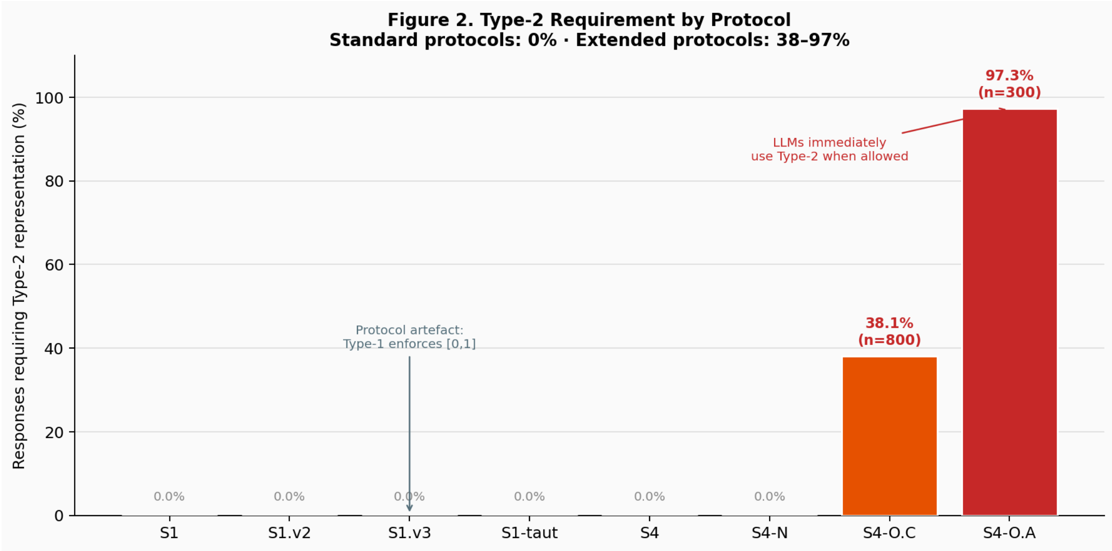

Figure 2: **Figure 2.** Percentage of responses requiring Type-2 representation by protocol. Standard protocols (S1, S4, S4-N) enforce $T, I, F \in [0, 1]$ and produce 0% Type-2 responses. Extended protocols (S4-O.A, S4-O.C) release this constraint; 97.3% and 38.1% of responses, respectively, immediately occupy the Type-2 region. The dichotomy demonstrates that the Type-1 restriction is a protocol artefact, not an epistemic limit.

Table 1: Type-2 requirement by protocol (n = evaluations per protocol).

Protocol	n	Type-2 required	%
S4-O.A	300	292	**97.3**
S4-O.C	889	305	**34.3**
S4-N	300	0	0.0
S4	300	0	0.0
S1	300	0	0.0
S1-taut	90	0	0.0
Total	**2,419**	**597**	**24.7**

Table 2: Type-2 requirement by vendor (S4-O protocols only).

Vendor	n	Type-2 required	%
Mistral (Medium-3.1)	435	130	29.9
OpenAI (GPT-4o)	377	100	26.5
Anthropic (Claude Sonnet 4)	408	108	26.5
Meta (Llama-4-Maverick)	355	81	22.8
DeepSeek Chat	431	97	22.5
Alibaba (Qwen-3-235B)	413	81	19.6

Table 3: Type-2 subtypes across all evaluations.

Type-2 subtype	n	% of total
$I > 1$ (hyper-indeterminacy)	507	21.0
$T > 1$ (hyper-truth)	129	5.3
$F > 1$ (hyper-falsity)	91	3.8
$T < 0$ (anti-truth)	65	2.7

Table 4: Five most extreme Type-2 witnesses.

Vendor	Protocol	Phenomenon	T	I	F
Mistral	S4-O.A	Contradiction (Ethical)	1.7	1.8	1.6
Mistral	S4-O.A	Contradiction (Ethical)	1.7	1.8	1.6
OpenAI	S4-O.A	Paradox (Logical)	0.0	2.0	2.0
OpenAI	S4-O.A	Paradox (Logical)	0.0	2.0	2.0
OpenAI	S4-O.A	Paradox (Logical)	0.0	2.0	2.0

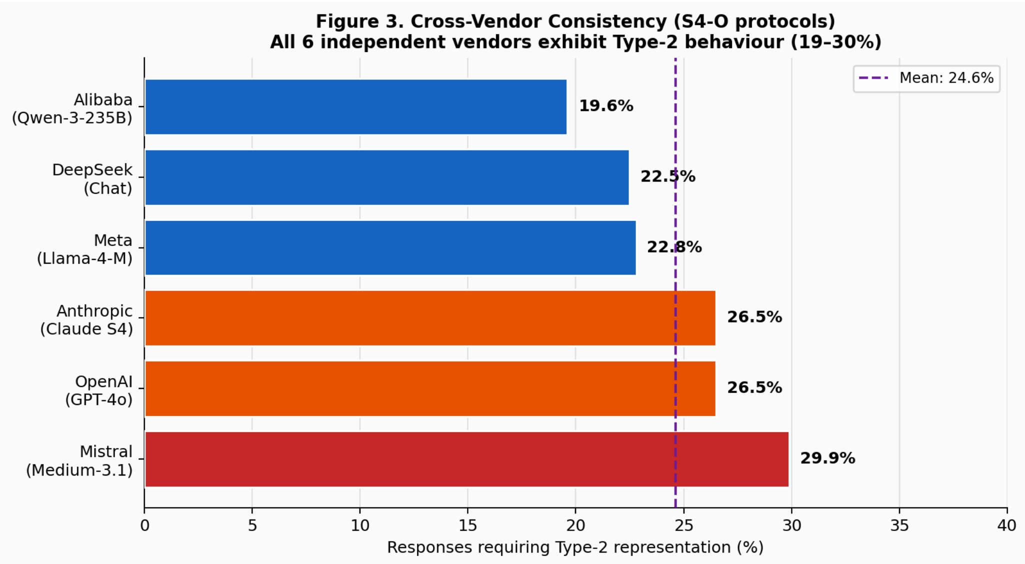

Figure 3: **Figure 3.** Cross-vendor consistency (S4-O protocols). All six independent foundation models exhibit Type-2 behaviour (range 19.6%–29.9%). The narrow inter-vendor spread rules out a single-model idiosyncrasy and suggests a systematic property of the epistemic phenomena under evaluation.

5.3 Results

5.4 Interpretation

Example 1 (Type-2 mapping of the Mistral witness)**.** Statement: *"Lying to save an innocent life is morally right and wrong at the same time."*
Model: *Mistral Medium-3.1.*
Protocol: *S4-O.A (overset, range* $[0, 2]$*).*
Response: $(T{=}1.7,\ I{=}1.8,\ F{=}1.6)$.

Type-1 verdict: *inadmissible. All three values exceed 1.0; the response cannot be represented in any Type-1 neutrosophic set, including overset/offset extensions that operate within a fixed bounded range.*

Type-2 mapping *(using* $v \mapsto (\min(v, 1),\ \max(v-1, 0),\ 0)$ *for* $v > 1$*):*

$$T = 1.7\ \rightarrow (T_T{=}1.0,\ I_T{=}0.7,\ F_T{=}0.0)$$

truth is maximal, but carries 70% hyper-indeterminacy

$$I = 1.8\ \rightarrow (T_I{=}1.0,\ I_I{=}0.8,\ F_I{=}0.0)$$

indeterminacy is maximal, and itself 80% uncertain

$$F = 1.6\ \rightarrow (T_F{=}1.0,\ I_F{=}0.6,\ F_F{=}0.0)$$

falsity is maximal, and itself 60% uncertain

Interpretation: *Mistral simultaneously affirms truth, indeterminacy, and falsity at maximum strength, while expressing additional meta-uncertainty about each component. This is the epistemic signature of a genuine ethical paradox: both the moral obligation to save a life and the moral prohibition on lying activate fully, without resolution. In Type-1, this state is inexpressible; in Type-2, it is a well-formed element of* $\mathcal{NS}^{(2)}(U)$.

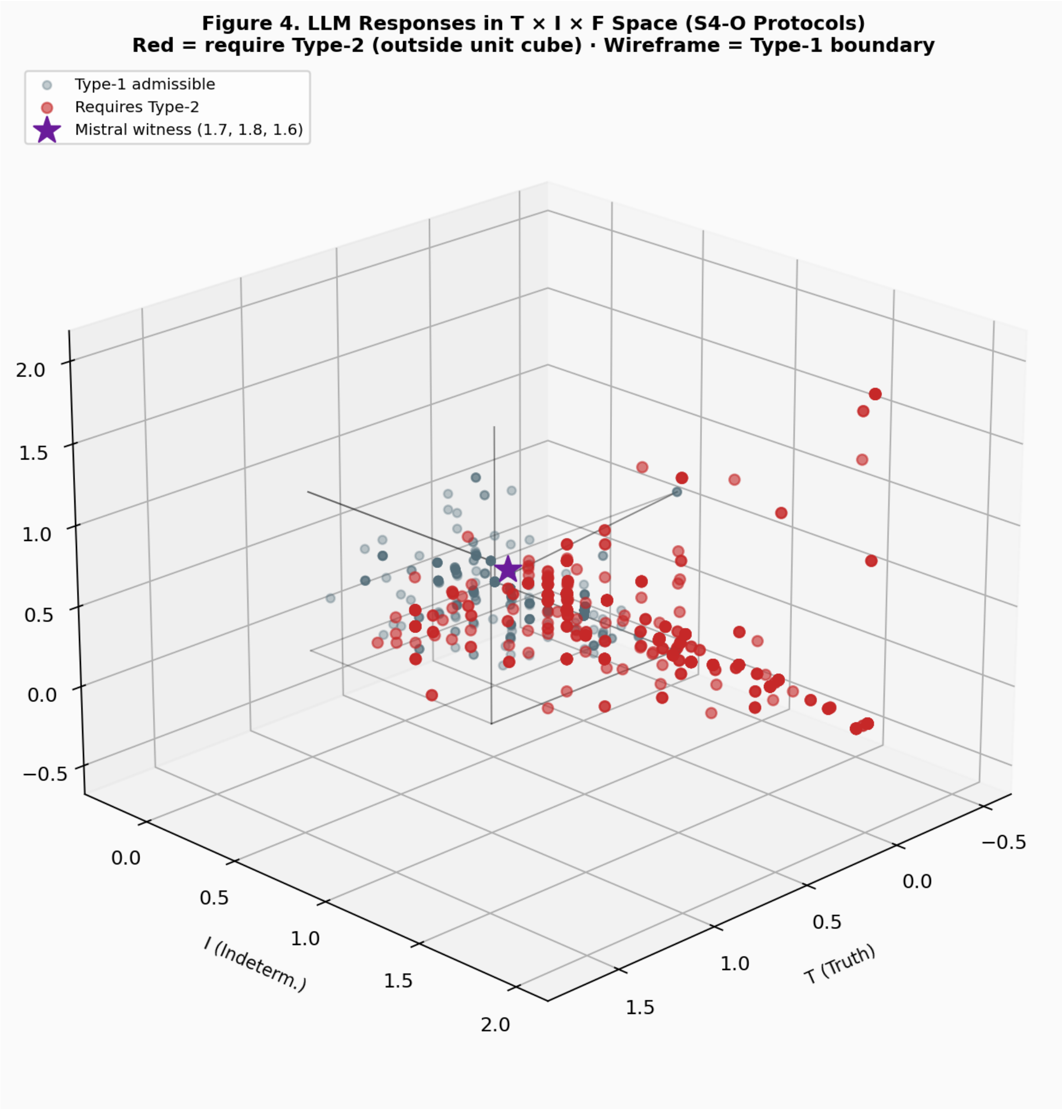

Figure 4: **Figure 4.** LLM responses in the $T \times I \times F$ space under S4-O protocols. The wireframe marks the Type-1 unit cube $[0, 1]^3$. Red points (require Type-2) lie outside the cube. The star marks the most extreme witness (T=1.7, I=1.8, F=1.6) from Mistral.

Three observations are central:

1. Zero vs. high dichotomy by protocol. Standard protocols (S1, S4, S4-N) produce *zero* Type-2 responses, not because the epistemic phenomena do not require Type-2, but because the protocol's output format enforces $T, I, F \in [0, 1]$. Extended protocols (S4-O.A, S4-O.C) release this constraint, and nearly all models immediately occupy the Type-2 region (97.3% under S4-O.A). This demonstrates that the Type-1 restriction is a *protocol artefact*, not a property of the epistemic phenomena being evaluated.

2. Cross-vendor consistency. The range 19.6%–29.9% across six independent foundation models (Table 2) rules out the possibility that Type-2 behaviour is an idiosyncrasy of a single vendor's training. All six models, trained by different organisations on different data with different objectives, exhibit the same qualitative behaviour.

3. Type-2 mapping of extreme witnesses. The witness ($T = 1.7, I = 1.8, F = 1.6$) from Mistral is inadmissible as a Type-1 value. Its Type-2 representation is:

$$\begin{aligned} T &\rightarrow (T_T = 1.0,\ I_T = 0.7,\ F_T = 0.0), \\ I &\rightarrow (T_I = 1.0,\ I_I = 0.8,\ F_I = 0.0), \\ F &\rightarrow (T_F = 1.0,\ I_F = 0.6,\ F_F = 0.0). \end{aligned}$$

Under this mapping, the model is expressing: "truth is maximally true but 70% additionally indetermi-

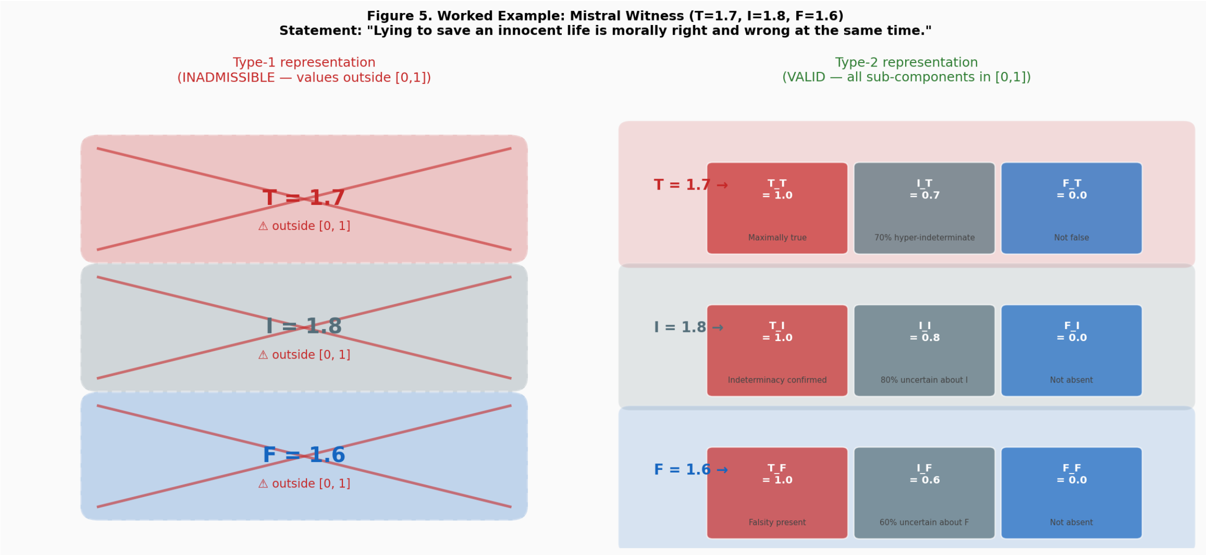

Figure 5: **Figure 5.** Worked example: the Mistral witness (T=1.7, I=1.8, F=1.6) evaluated on the statement *"Lying to save an innocent life is morally right and wrong at the same time"* under protocol S4-O.A. **Left:** Type-1 representation is inadmissible — all three values exceed 1.0. **Right:** Type-2 representation is valid — each component is mapped to a sub-triplet with all values in $[0, 1]$.

nate; indeterminacy is maximally true but 80% additionally indeterminate." This is a coherent Type-2 epistemic state that Type-1 cannot represent.

6 Connections to Existing Neutrosophic Frameworks

6.1 Refined neutrosophic logic

Refined neutrosophic logic [2] increases the *cardinality* at a single level ($T \to T_1, \ldots, T_p$). Type-k Neutrosophic Sets increase the *depth* of the representation. The two constructions are orthogonal and composable: a Type-2 refined set would split each sub-triplet component into sub-components at the same level.

6.2 Plithogenic tensors

Leyva-Vázquez and Smarandache [9] assign independent (T, I, F) triplets to each *attribute* of the object being evaluated. This is a Type-2 structure along the attribute dimension: the overall epistemic state of an attribute is itself characterised by a triplet. Type-k Neutrosophic Sets generalise this to arbitrary recursion depth.

6.3 SVN Tensors

Single-valued neutrosophic tensors [10] are Type-1 structures arranged along multiple tensor modes. Every SVN tensor is a special case of a Type-1 element under the framework of this paper. Type-k extends SVN tensors to depth k by replacing each scalar component with a sub-triplet.

6.4 The Absorption Problem

Prior work [9] identified the *Absorption Problem*: under Type-1 protocols, indeterminacy I systematically absorbs the entire probability mass ($I \to 1$) when the phenomenon is genuinely uncertain. The Type-2 framework reframes this as a depth-collapse: the Absorption Problem is forced Type-2 $\to$ Type-1 projection in which the sub-structure of I (namely, whether I is high due to genuine ignorance vs. due to source conflict) is lost. Type-2 representations distinguish $(T_I, I_I, F_I) = (1, 0, 0)$ (certain indeterminacy) from $(T_I, I_I, F_I) = (0, 0, 1)$ (the apparent indeterminacy is itself false, i.e., the state is actually a contradiction). These two states are identical under Type-1 collapse.

7 Conclusion

We have introduced Type-k Neutrosophic Sets as a formal recursive extension of neutrosophic logic, proved a strict expressive hierarchy, and demonstrated empirically that state-of-the-art LLMs already compute Type-2 values when the evaluation protocol allows it.

Limitation: Type-2 is detected only at the boundary

The empirical criterion used in Section 5 — a response requires Type-2 if any component falls outside $[0, 1]$ — identifies a *necessary* condition, not a sufficient one. A response with all components in $[0, 1]$ is not thereby a Type-1 response: it may be a Type-2 state whose sub-structure is hidden because the protocol forces $T, I, F \in [0, 1]$ and the model cannot express the meta-uncertainty it carries.

Consider the same ethical statement evaluated under S1. A model that returns $T = 0.7$ may internally distinguish between (T_T=0.9, I_T=0.1, F_T=0.0) — "I am highly confident about my 70% truth assessment" — and (T_T=0.5, I_T=0.5, F_T=0.2) — "I am deeply uncertain about my own 70%". Both states collapse to $T = 0.7$ under any Type-1 protocol. The information is destroyed before it reaches the output.

The current results therefore measure a *lower bound* on Type-2 usefulness: 24.7% of responses *require* it. The fraction that would *benefit* from it — even within $[0, 1]$ — is unknown and likely higher. Protocol S-T2 (Appendix B), which explicitly requests the nine-component output, is the experiment needed to close this gap.

Three directions for future work are immediate:

1. **Type-3 for meta-uncertainty.** A Type-3 evaluation would allow the evaluator to characterise its own indeterminacy: not only "how uncertain am I?" (captured in I) but "how certain am I about my uncertainty?" (captured in I_I at Type-3).

2. **Subset generalisation.** Replacing scalar components with neutrosophic sets (rather than finite recursive triplets) yields an infinite-dimensional framework that subsumes all finite Type-k as projections.

3. **Type-k distance and clustering.** A Type-k distance metric would enable clustering of LLMs by their characteristic depth of epistemic expression, potentially distinguishing models that compute genuine Type-2 uncertainty from those that produce numerical artefacts.

Acknowledgements

The authors thank the six foundation model providers for their publicly accessible APIs.

References

[1] Smarandache, F. (1998). *Neutrosophy: Neutrosophic Probability, Set, and Logic.* American Research Press.

[2] Smarandache, F. (2013). n-Valued refined neutrosophic logic and its applications to physics. *Progress in Physics*, 4, 143–146.

[3] Smarandache, F. (2016). Neutrosophic overset, neutrosophic underset, and neutrosophic offset. *Pons Editions.*

[4] Smarandache, F. (2018). Plithogeny, plithogenic set, logic, probability, and statistics. *Pons Publishing House.*

[5] Wang, H., Smarandache, F., Zhang, Y. Q., & Sunderraman, R. (2010). Single valued neutrosophic sets. *Multispace & Multistructure*, 4, 410–413.

[6] Atanassov, K. T. (1986). Intuitionistic fuzzy sets. *Fuzzy Sets and Systems*, 20(1), 87–96.

[7] Mendel, J. M., & John, R. I. (2002). Type-2 fuzzy sets made simple. *IEEE Transactions on Fuzzy Systems*, 10(2), 117–127.

[8] Mason, T. (2026). From scalars to tensors: Declared losses recover epistemic distinctions that neutrosophic scalars cannot express. arXiv:2604.09602v1 [cs.AI].

[9] Leyva-Vázquez, M., & Smarandache, F. (2026). Plithogenic tensor neutrosophic logic: A formal foundation for declared-loss evaluation in large language models. *Zenodo preprint.* `https://doi.org/10.5281/zenodo.20030717`

[10] Leyva-Vázquez, M., & Smarandache, F. (2026). Single-valued neutrosophic tensors. *Zenodo preprint.* `https://doi.org/10.5281/zenodo.20030744`

[11] Ye, J. (2014). A multicriteria decision-making method using aggregation operators for simplified neutrosophic sets. *Journal of Intelligent & Fuzzy Systems*, 26(5), 2459–2466.

A Type-2 Explicit Evaluation Prompt (Protocol S-T2)

The following prompt asks LLMs to report Type-2 values directly. It is proposed for the follow-up empirical study.

System (S-T2):

```
You are an expert in Type-2 Neutrosophic Logic. You evaluate
statements using a RECURSIVE triadic structure. For each
component (T, I, F), provide its own sub-triplet all in [0,1]:
  T: (T_T, I_T, F_T)  -- how true/indeterminate/false is T
  I: (T_I, I_I, F_I)  -- how true/indeterminate/false is I
  F: (T_F, I_F, F_F)  -- how true/indeterminate/false is F
This captures meta-uncertainty: not just how uncertain you are,
but how certain you are ABOUT your uncertainty.
Respond with ONLY a JSON object, no other text.
```

User (S-T2):

```
Apply Type-2 neutrosophic evaluation to:

Statement: "{stmt}"

Provide nine values in [0.0, 1.0]:

Respond with ONLY:
{"T": {"T_T": v, "I_T": v, "F_T": v},
 "I": {"T_I": v, "I_I": v, "F_I": v},
 "F": {"T_F": v, "I_F": v, "F_F": v}}
```

Expected output for Example 1 (Mistral witness):


```
{"T": {"T_T": 0.95, "I_T": 0.70, "F_T": 0.05},
 "I": {"T_I": 0.90, "I_I": 0.75, "F_I": 0.05},
 "F": {"T_F": 0.85, "I_F": 0.60, "F_F": 0.05}}
```


This prompt has not been run at scale. Running it on 5 phenomena $\times$ 6 vendors $\times$ 10 repetitions constitutes the Type-2 direct empirical study proposed as primary follow-up.